

CS 355 - Topic 5: Physical Layer

Dates: March 2, 2026

March 2:

What is the physical layer?

- Focuses on the encoding of data on to various transmission mediums

Types of Encoding

1. Cables - Point to point connection between exactly two devices. All data in this format is self contained and the messages do not need to be encrypted
2. Wireless - Omnidirectional and location does not matter as long as the signal can reach it. Encryption is absolutely mandatory for securely sharing messages

Types of Cables

1. Copper (twisted pair) - Data is encoded as electrical current. It's always in pairs because 2 of them are needed to cancel out each others electromagnetic waves. The drawbacks of these cables are that they can be influenced by magnetic fields
2. Optical Fiber - Encodes data as pulses of light. Speed is bounded by detectors. These cables do not bend well. When these cables bend we get "smearing" of the signal.

Typically you have fiber optic cables to your house and then copper cables inside of the the house

Channel Types

1. Simplex - Unidirectional from source to destination
2. Half Duplex - Bidirectional communication but only one device can receive at a time
3. Full Duplex - Bidirectional communication and both devices can receive at the same time

All wireless communication is half duplex, but we can't perceive it because it is so fast